Case Study - Telecom DWH

A telecom company has contacted you in your capacity as an ETL Developer/Consultant to design and implement a data loading process according to the following requirements:

- **Source System**

The company has a system that generates a CSV file every 5 minutes. This file contains the core data of various customer transactions performed within a specific time window.

- **CSV File Structure**

The company provided the following structure describing the data stored in the CSV file:

Column Name	Data Type	Length	Is Nullable	Sample
ID	Int	—	False	156
IMSI	String	9	False	310120265
IMEI	String	14	True	490154203237518
CELL	Int	—	False	123
LAC	Int	—	False	12
EVENT_TYPE	String	1	True	1
EVENT_TS	Timestamp	—	False	16/1/2020 7:45:43

- **Required Data Processing**

Before Loading into the Database The following transformation and validation rules must be applied before storing the data:

Column Name	Mapping Rules	Target Model
ID	As-is	Transaction_id
IMSI	As-is, reject the record if null	IMSI
IMSI	Join with IMSI reference table to get subscriber_id; if not found, add -99999	subscriber_id
IMEI	Substring first 8 characters; if null or length < 14, set -99999	TAC
IMEI	Substring last 6 characters; if null or length < 14, set -99999	SNR
IMEI	As-is	IMEI
CELL	As-is, reject the record if null	CELL
LAC	As-is, reject the record if null	LAC
EVENT_TYPE	As-is	EVENT_TYPE
EVENT_TS	Validate timestamp format as datetime; reject the record if invalid or null	EVENT_TS

- **Rejected Records Handling**

Rejected records must be stored in a separate table and The original CSV file name must also be recorded along with each rejected record.

- **File Handling After Processing**

After successfully loading the valid data into the database move the processed CSV file to another folder (Archive/Processed folder).